



# Girraween High School

## 2024 Year 12 Mathematics Extension 2 Task 1, Dec-2023

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### General Instructions

- Working Time - 90 minutes
- Write using black or blue pen only
- Calculators approved by NESA may be used
- For questions in Section II, Show relevant mathematics reasoning and/ or calculations

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**Total Marks: 95**

### Section I - 5 marks (page 1)

- Attempt questions 1-5
- Allow about 10 minutes for this section

### Section II - 87 marks (pages 3-6)

- Attempt questions 6-11
- Allow about 80 minutes for this section

**Question 1** (1 mark)

Which of the following gives  $\sqrt{3} + i$  in Modulus-argument form.

A.  $2cis\left(\frac{-\pi}{6}\right)$

B.  $2cis\left(\frac{-2\pi}{3}\right)$

C.  $2cis\left(\frac{\pi}{6}\right)$

D.  $2cis\left(\frac{2\pi}{3}\right)$

**Question 2** (1 mark)

Which is the simplified form of  $i^{2011}$  ?

A.  $i$

B.  $-1$

C.  $-i$

D.  $1$

**Question 3** (1 mark)

The relation  $(z + 2)(\bar{z} + 2) = 4$ , when graphed on an Argand diagram, would be

A. a circle of radius 4 with centre at  $(-2,0)$

B. a circle of radius 4 with centre at  $(2,0)$

C. a circle of radius 2 with centre at  $(-2,0)$

D. a circle of radius 2 with centre at  $(2,0)$

**Question 4** (1 mark)

What are the values of real numbers  $p$  and  $q$  such that  $1 - i$  is a root of the equation

$P(z) = z^3 + pz + q$ ?

A.  $p = -2$  and  $q = -4$

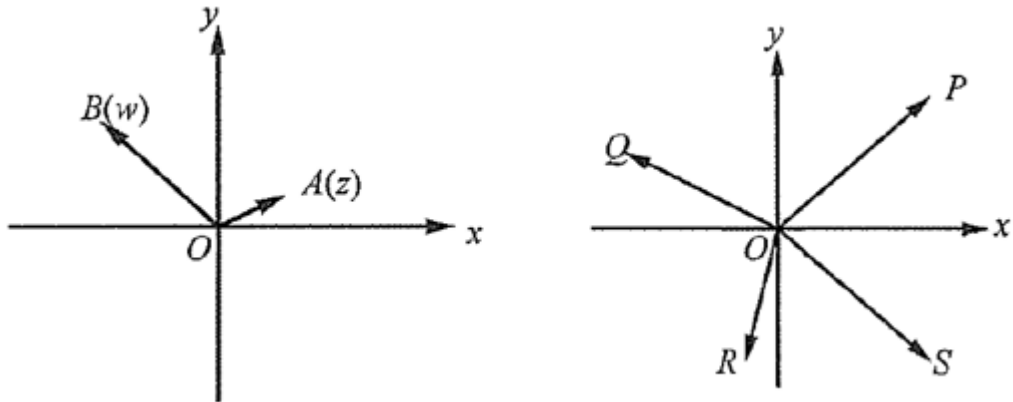
B.  $p = -2$  and  $q = 4$

C.  $p = 2$  and  $q = -4$

D.  $p = 2$  and  $q = 4$

**Question 5** (1 mark)

The vectors  $OA$  and  $OB$  represent the complex numbers  $z$  and  $w$  respectively.



which vector represents the complex number  $\overline{(z - iw)}$

- A.  $OP$
- B.  $OQ$
- C.  $OR$
- D.  $OS$

## Mathematics Extension 2

### Section II

87 Marks

Attempt Questions 6-11

Allow about 80 minutes for this section

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#### Question 6 (13 marks)

- (a) If  $z_1 = 3 - 2i$  and  $z_2 = 3 - 4i$ , express the following in the form  $a + ib$  where  $a$  and  $b$  are real.

i.  $z_1 - z_2$  [1]

ii.  $z_2^2$  [1]

iii.  $z_1 \overline{z_2}$  [1]

- (b) i. Find the values of the real numbers  $x$  and  $y$  such that  $(x + iy)^2 = -24 + 10i$  [3]

ii. Hence, solve the quadratic equation  $z^2 - (3 + i)z + 8 - i = 0$  [2]

- (c) Find the modulus of  $\frac{(2 - i)^8}{(2 + i)^6}$  [3]

- (d) Simplify  $\left(e^{\frac{-i\pi}{3}}\right)^2 (1 + i\sqrt{3})^3$ , writing your answer in the form  $x + iy$  where  $x, y$  are real [2]

**Question 7** (17 marks)

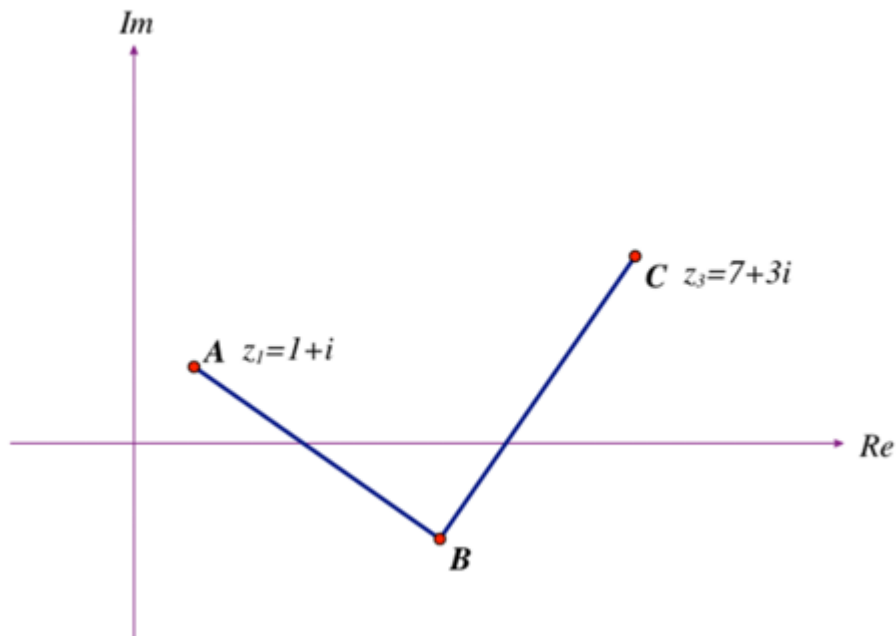
- (a) Sketch the region in the Argand diagram containing the points  $z$  for which
- i.  $|\arg z| \leq \frac{\pi}{6}$  [1]
  - ii.  $z + \bar{z} < 4$  [1]
  - iii.  $2 < |z| < 4$  [2]
  - iv.  $\arg\left(\frac{z+1}{z-3}\right) = \frac{\pi}{3}$  [2]
- (b) On an Argand diagram, sketch the region satisfied by both inequalities: [3]
- $$|z+1| \leq |z-i| \text{ and } \operatorname{Im}(z) < 2$$
- (c) i. On an Argand diagram shade the region containing all points representing complex numbers  $z$  such that both  $|z-1| \leq 1$  and  $0 \leq \operatorname{Arg}(z-1) \leq \frac{\pi}{4}$ . [2]
- ii. If the point  $P$  in this region represents the complex number  $z = z_1$  which has the largest principal argument, express  $z_1$  in modulus/argument form. [2]
- (d)  $|z| = 1$  and  $z = \cos\theta + i\sin\theta$ , where  $-\pi < \theta \leq \pi$ .
- i. Show that  $1+z = 2\cos\frac{\theta}{2} \left( \cos\frac{\theta}{2} + i\sin\frac{\theta}{2} \right)$  [2]
  - ii.  $z_1$  and  $z_2$  are complex numbers such that  $|z_1| = |z_2| = 1$ . If  $z_1$  and  $z_2$  have arguments  $\alpha$  and  $\beta$  respectively, where  $-\pi < \alpha \leq \pi$  and  $-\pi < \beta \leq \pi$ , show that  $\frac{z_1 + z_1 z_2}{z_1 + 1}$  has modulus  $\left( \frac{\cos\frac{\beta}{2}}{\cos\frac{\alpha}{2}} \right)$  and argument  $\left( \frac{\alpha + \beta}{2} \right)$  [2]

**Question 8** (13 marks)

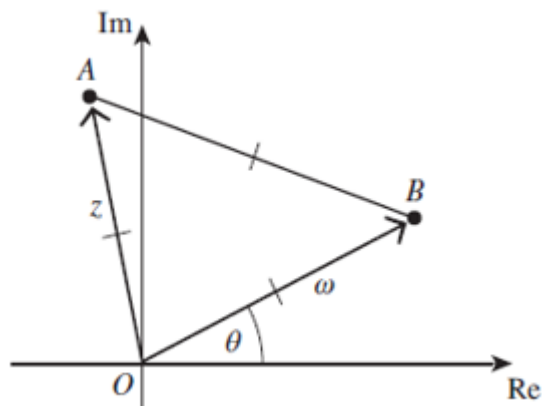
- (a) i. Realise the denominator of  $\frac{\sqrt{3}-i}{1-i}$  and express your answer in cartesian form. [2]
- ii. Show that  $\sqrt{3}-i$  and  $1-i$  are same as  $2\operatorname{cis}\left(-\frac{\pi}{6}\right)$  and  $\sqrt{2}\operatorname{cis}\left(-\frac{\pi}{4}\right)$  respectively. [2]
- iii. Hence find the exact value of  $\sin\frac{\pi}{12}$  [2]
- (b) The polynomial  $P(z) = z^4 - 10z^3 + cz^2 + dz + 169$  has two zeros  $\alpha = a + ib$  and  $\beta = b + ia$ , where  $a, b, c, d$  are real and  $a \neq b$ .
- i. Express the other two zeros of  $P(z)$  in terms of  $a$  and  $b$ , justifying your answer. [1]
  - ii. Find the values of  $a$  and  $b$ , given that they both are integers. [2]
- (c) Find  $\alpha$  and  $\beta$  given that  $z^3 + 6z - 4\sqrt{2}i = (z - \alpha)^2(z - \beta)$  [4]

**Question 9** (15 marks)

- (a) The points  $A$  and  $C$  represent the complex numbers  $z_1 = 1 + i$  and  $z_3 = 7 + 3i$  and  $B$  represents the complex number  $Z_2$ . Find the complex number  $Z_2$  in the form  $a + ib$  where  $a$  and  $b$  are real, such that  $\triangle ABC$  is isosceles and right-angled at  $B$ . [4]



- (b) Let  $z = \overrightarrow{OA}$  and  $\omega = \overrightarrow{OB}$  represent the two sides of an equilateral triangle  $OAB$ , as shown.



- i. Find the modulus and argument of  $\frac{z}{\omega}$  [2]
- ii. Hence, find  $z^3 + \omega^3$ . [2]

**Question 9 continued onto next page...**

- (c) If  $z = \frac{1+i}{1-i}$  and  $\omega = \frac{\sqrt{2}}{1-i}$
- Calculate the modulus of  $z$  and  $\omega$  [2]
  - Calculate argument of  $z$  and  $\omega$  [2]

On an Argand diagram the points  $A$ ,  $B$  and  $C$  represent the numbers  $z$ ,  $z + \omega$  and  $\omega$  respectively.

- By considering the quadrilateral  $OABC$  and the argument of  $z + \omega$ , show that  $\tan \frac{3\pi}{8} = 1 + \sqrt{2}$ . [3]

**Question 10** (16 marks)

- Show that if  $z = \cos \theta + i \sin \theta$  then  $z^n + \frac{1}{z^n} = 2 \cos n\theta$  [3]
  - Hence, express  $\cos^5 \theta$  in terms of  $\cos 5\theta$ ,  $\cos 3\theta$ ,  $\cos \theta$  [4]
- Use De Moivre's theorem to find expressions for  $\cos 5\theta$  and  $\sin 5\theta$ . You may leave your answers in terms of mixed powers of  $\cos \theta$  and  $\sin \theta$ . [1]
  - If  $t = \tan \theta$  then show that  $\tan 5\theta = \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}$  [1]
  - Use the result from (ii) and an appropriate substitution to show that  $\tan \frac{\pi}{5} = \sqrt{5 - 2\sqrt{5}}$ . [3]
- Show that  $(1 + i \tan \theta)^n + (1 - i \tan \theta)^n = \frac{2 \cos n\theta}{\cos^n \theta}$ , where  $n > 0$ . [2]
  - Hence, show that the roots of  $(1 + z)^2 + (1 - z)^2 = 0$  are  $z = \pm i \tan \frac{\pi}{4}$  when  $z$  is purely imaginary. [2]

**Question 11** (13 marks)

- If  $\omega$  is the complex root of  $z^5 - 1 = 0$  with the smallest positive argument, show that  $\omega^2$ ,  $\omega^3$  and  $\omega^4$  are also roots of  $z^5 - 1 = 0$ . [2]
  - Show  $\omega + \omega^2 + \omega^3 + \omega^4 = -1$  [2]
- By matching pair of conjugate roots, resolve  $z^5 - 1 = 0$  into real linear and quadratic factors. [2]
  - Hence, resolve  $z^4 + z^3 + z^2 + z + 1$  into real quadratic factors [1]
  - Hence, show that  $2 \cos 2\theta + 2 \cos \theta + 1 = \left(2 \cos \theta - 2 \cos \frac{2\pi}{5}\right) \left(2 \cos \theta + 2 \cos \frac{2\pi}{5}\right)$  [3]
  - Hence, by substituting in an appropriate value for  $\theta$  show that  $\left(1 - 2 \cos \frac{2\pi}{5}\right) \left(1 + 2 \cos \frac{\pi}{5}\right) = 1$  [3]

**End of Task 1**

Task: HSC Assessment Task 1

Year: 2024

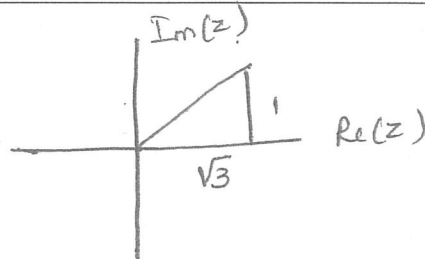
Suggested Solutions

Question 1

$$z = \sqrt{3} + i$$

$$\arg(z) = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

$$= 2 \operatorname{cis} \frac{\pi}{6}$$



(C)

Question 2

$$i^{2011} = i^3 + (502 \times 4)$$

$$= i^3$$

$$= -i$$

(C)

Marker's Feedback

## Suggested Solutions

Question 3

$$(z+2)(\bar{z}+2) = (z\bar{z} + 2z + 2\bar{z} + 4) \\ = z\bar{z} + 2(z + \bar{z}) + 4$$

$$(z+2)(\bar{z}+2) = 4; \text{ hence } z\bar{z} + 2(z + \bar{z}) = 0$$

$$\text{let } z = x + iy$$

$$z\bar{z} + 2(z + \bar{z}) = 0$$

$$(x+iy)(x-iy) + 2(x+iy + x-iy) = 0$$

$$x^2 + y^2 + 4x = 0$$

$$(x+2)^2 + y^2 = 2^2$$

$\therefore$  circle with centre  $(-2, 0)$  and radius 2.

(C)

Marker's Feedback

## Suggested Solutions

Question 4

All the coefficients of  $z^3 + 0z^2 + pz + q$  are real.  
 Then any complex roots occur in conjugate pairs.  
 Since  $1-i$  is a root then  $1+i$  is a root.

Sum of the roots.

$$(1-i) + (1+i) + \alpha = -\frac{b}{a}$$

$$\alpha + 2 = 0$$

$$\alpha = -2$$

Product of the roots

$$(1-i)(1+i)(-2) = -\frac{d}{a}$$

$$(1+1)(-2) = -q$$

$$\boxed{q = 4}$$

Sum of the roots two at a time

$$(1-i)(1+i) + (1-i)(-2) + (1+i)(-2) = \frac{c}{a}$$

$$2 + (-2) + 2i + (-2) - 2i = \frac{p}{1}$$

$$\boxed{\therefore p = -2}$$

(B)

Marker's Feedback

## Suggested Solutions

Question 5 : Answer (D).

Question 6

(a) Given  $z_1 = 3 - 2i$  ;  $z_2 = 3 - 4i$

(i),  $z_1 z_2 = (3 - 2i)(3 - 4i)$   
 $= 2i$

(ii),  $(z_2)^2 = (3 - 4i)^2$   
 $= 9 - 24i + 16i^2$   
 $= -7 - 24i$

(iii),  $z_1 \bar{z}_2 = (3 - 2i)(3 + 4i)$   
 $= 9 + 12i - 6i + 8$   
 $= 17 + 6i$

## Marker's Feedback

Generally well done — just a few silly mistakes.

## Suggested Solutions

Question 6 b, i,

$$(x+iy)^2 = -24 + 10i$$

$$x^2 - y^2 + 2ixy = -24 + 10i$$

Equating real and imaginary parts

$$x^2 - y^2 = -24 \quad (1)$$

$$xy = 5 \Rightarrow y = \frac{5}{x} \quad (2)$$

Sub (2) in (1)

$$x^2 - \left(\frac{5}{x}\right)^2 = -24$$

$$x^4 - 25 = -24x^2$$

$$x^4 + 24x^2 - 25 = 0$$

$$(x^2 + 25)(x^2 - 1) = 0$$

$x \neq \pm 5i$   
( $x$  real).

$$x = \pm 1. \quad \text{Sub. } x = 1 \text{ in (2)} \quad \left| \quad \text{Sub. } x = -1 \text{ in (2)} \right.$$

$$y = \frac{5}{1} = 5.$$

$$y = \frac{5}{-1} = -5.$$

## Marker's Feedback

Generally well done — EXCEPT

→ A small number of students used an INCORRECT method [described below]:

$$x^2 + y^2 = 26 \Rightarrow \text{True as } |-24 + 10i| = x^2 + y^2$$

$$\underline{x^2 - y^2 = 24 \quad (2)}$$

(1) + (2) gets  $x = \pm 5$

& (1) - (2) gets  $y = \pm 1$

BUT IF YOU USE THIS METHOD,  
YOU DON'T KNOW IF  $y = 1$  or  $-1$  WHEN  $x = 5$ .  
DO NOT DO THIS!

## Suggested Solutions

Question 6b.i, continued

$$x=1, y=5 \text{ or } x=-1, y=-5$$

The complex square roots are  $1+5i$  and  $-1-5i$

$$\text{b.ii, } z^2 - (3+i)z + 8-i = 0$$

$$\Delta = 9+6i -1-32+4i$$

$$= -24+10i$$

$$z = \frac{3+i \pm (1+5i)}{2} \text{ from i}$$

$$z = 2+3i \text{ or } 1-2i$$

## Marker's Feedback

(6)(b)(ii) Generally well done.

## Suggested Solutions

Question 6c

modulus of  $\frac{(2-i)^8}{(2+i)^6}$

$$|2-i| = \sqrt{5}, \quad |2+i| = \sqrt{5}$$

$$\therefore \left| \frac{(2-i)^8}{(2+i)^6} \right| = \frac{|(2-i)|^8}{|(2+i)|^6} = \frac{\sqrt{5}^8}{\sqrt{5}^6} = 5$$

6d

$$\begin{aligned} \left( e^{-i\pi/3} \right)^2 (1+i\sqrt{3})^2 &= \left( e^{-i2\pi/3} \right) \cdot \left( 2e^{i\pi/3} \right)^3 \\ &= e^{-i2\pi/3} \cdot 8e^{i\pi} \\ &= 8e^{i\pi/3} \end{aligned}$$

$$\begin{aligned} &= 4 + 4\sqrt{3}i \\ \text{Note: Could also do: } &\left( \cos \frac{\pi}{3} - i \sin \frac{\pi}{3} \right)^2 \times \left( 2 \cos \frac{\pi}{3} + i 2 \sin \frac{\pi}{3} \right)^3 \\ &= e^{i\pi} \times 8e^{-i2\pi/3} \\ &\text{etc.} \end{aligned}$$

## Marker's Feedback

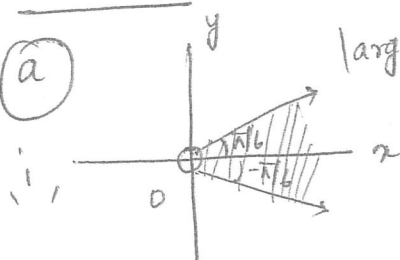
Generally well done, apart from silly mistakes.  
The 2 most common were  $[2e^{i\pi/3}]^3 = 2e^{i\pi}$   
[they forgot to cube 2].

and forgetting that  $-8 = 8e^{i\pi}$  and you  
add to add  $-\frac{2\pi i}{3}$  and  $\pi i$ .

Suggested Solutions

Question 7

(a)

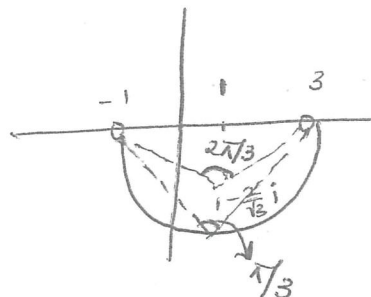


$$|\arg z| \leq \pi/6$$

Some students forgot to put open circle at origin.

iv,

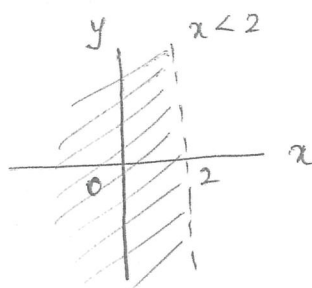
$$\arg \left( \frac{z+1}{z-3} \right) = \frac{\pi}{3}$$



(ii)

$$z + \bar{z} \leq 4$$

$$z + \bar{z} = 2\operatorname{Re}(z) < 4$$

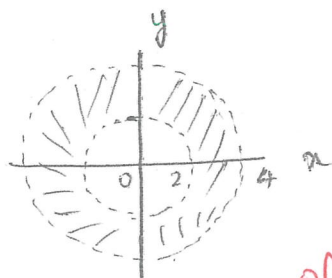


$$x < 2$$

done well

(iii)

$$2 < |z| < 4$$



writing radii of circles as 4 and 16

Some students drew the arc above x axis; not showing the angle  $\frac{\pi}{3}$

Marker's Feedback

Question 16  $|z+1| \leq |z-i|$  and  $\text{Im}(z) < 2$

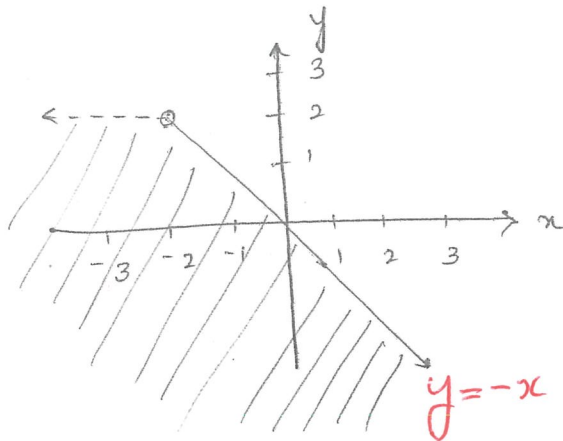
let  $z = x+iy$

$$|x+iy+1| \leq |x+iy-i|$$

$$|x+1+iy| \leq |x+i(y-1)|$$

$$(x+1)^2 + y^2 \leq x^2 + (y-1)^2$$

$$y \leq -x$$



\* Many are not writing the equation  $y = -x$ .

\* drawing  $y = -x$  as broken line

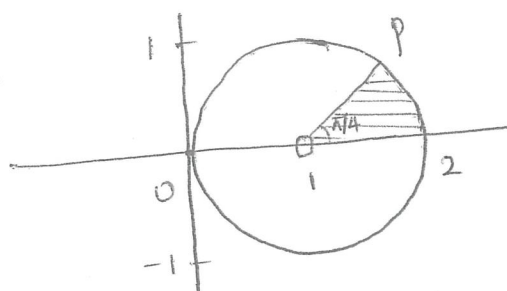
Marker's Feedback

Suggested Solutions

Question 7C(i)

Sketch and shade  $|z-1| \leq 1$  and  $0 \leq \arg(z-1) \leq \frac{\pi}{4}$

$$|z-1| \leq 1 \Rightarrow (x-1)^2 + y^2 = 1$$



no open circle at (1,0)

7C(ii)

The arc bounding the region subtends an angle  $\frac{\pi}{4}$  at the circle centre and hence an angle  $\frac{\pi}{8}$  at O on the circumference

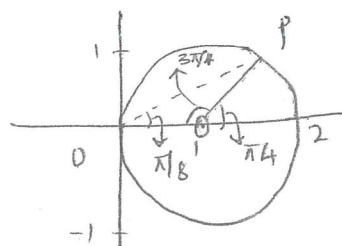
$$\therefore \arg(z_1) = \frac{\pi}{8}$$

Using the cosine rule:

$$OP^2 = 1^2 + 1^2 - 2 \cos \frac{3\pi}{4}$$

$$OP^2 = 2(1 - \cos \frac{3\pi}{4}) = 4 \sin^2 \frac{3\pi}{8}$$

$$\therefore OP = 2 \sin \frac{3\pi}{8} ; \text{ Hence } z_1 = 2 \sin \frac{3\pi}{8} \left( \cos \frac{\pi}{8} + i \sin \frac{\pi}{8} \right) = \sqrt{2} + i \sqrt{2} \left( \cos \frac{\pi}{8} + i \sin \frac{\pi}{8} \right)$$



Marker's Feedback

Students have difficulty with 7C(ii)  
Very few got the correct answers.

## Suggested Solutions

Question 7d i,

$$1+z = 1 + (\cos\theta + i\sin\theta)$$

$$= \left(1 + \cos\frac{2\theta}{2}\right) + \left(i\sin\frac{2\theta}{2}\right)$$

$$= 2\cos^2\frac{\theta}{2} + 2i\sin\frac{\theta}{2}\cos\frac{\theta}{2}$$

$$= 2\cos\frac{\theta}{2} \left(\cos\frac{\theta}{2} + i\sin\frac{\theta}{2}\right)$$

or

$$2\cos\frac{\theta}{2} \operatorname{cis}\frac{\theta}{2}.$$

done well.

## Marker's Feedback

## Suggested Solutions

Question

7dii,

$$1+z_1 = 2\cos\frac{\alpha}{2} \operatorname{cis}\frac{\alpha}{2} \quad ; \quad 1+z_2 = 2\cos\frac{\beta}{2} \operatorname{cis}\frac{\beta}{2}$$

$$\left| \frac{z_1(1+z_2)}{1+z_1} \right| = \left| \frac{\operatorname{cis}\alpha \left( 2\cos\frac{\beta}{2} \operatorname{cis}\frac{\beta}{2} \right)}{2\cos\frac{\alpha}{2} \operatorname{cis}\frac{\alpha}{2}} \right|$$

$$= \left| \frac{\operatorname{cis}\left(\alpha + \frac{\beta}{2}\right) \cdot 2\cos\frac{\beta}{2}}{\operatorname{cis}\frac{\alpha}{2} \cdot 2\cos\frac{\alpha}{2}} \right|$$

Now:

$$\left| \frac{z_1(1+z_2)}{1+z_1} \right| = \frac{\cos\frac{\beta}{2}}{\cos\frac{\alpha}{2}}$$

$$\begin{aligned} \operatorname{Arg}\left(\frac{z_1(1+z_2)}{1+z_1}\right) &= \frac{\alpha+\beta}{2} - \frac{\alpha}{2} \\ &= \frac{\alpha+\beta}{2} \end{aligned}$$

## Marker's Feedback

Many found the above questions hard.

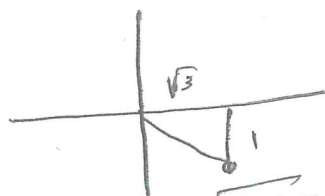
## Suggested Solutions

Question 8a, i,

$$\frac{\sqrt{3} - i}{1 - i} \times \frac{(1 + i)}{(1 + i)}$$

$$= \frac{\sqrt{3} + i\sqrt{3} - i + 1}{2}$$

$$= \frac{\sqrt{3} + 1}{2} + i \frac{(\sqrt{3} - 1)}{2}$$

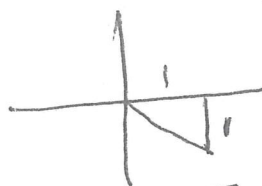
ii,  $\sqrt{3} - i$ 

$$\text{modulus} = \sqrt{3+1} = 2$$

$$\arg = -\tan^{-1}\left(\frac{\sqrt{3}}{1}\right)$$

$$= -\frac{\pi}{6}$$

$$\therefore \sqrt{3} - i = 2 \operatorname{cis}\left(-\frac{\pi}{6}\right)$$

 $1 - i$ 

$$\text{modulus} = \sqrt{2}$$

$$\arg(1 - i) = \tan^{-1}(1)$$

$$= -\frac{\pi}{4}$$

$$\therefore 1 - i = \sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)$$

## Marker's Feedback

8a, i, &amp; ii,

Mostly done well. Students have shown good understanding of Cartesian form & Mod-Arg form

## Suggested Solutions

Question 8a(iii),

$$\frac{\sqrt{3}-i}{1-i} = \frac{2 \operatorname{cis}\left(-\frac{\pi}{6}\right)}{\sqrt{2} \operatorname{cis}\left(-\frac{\pi}{4}\right)}$$

$$= \sqrt{2} \operatorname{cis}\left(\frac{\pi}{12}\right) = \sqrt{2} \left( \cos \frac{\pi}{12} + i \sin \frac{\pi}{12} \right)$$

Equating imaginary parts of (i) &amp; (iii),

$$\sqrt{2} \sin \frac{\pi}{12} = \frac{\sqrt{3}-1}{2}$$

$$\therefore \sin \frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}}$$

## Marker's Feedback

Mostly done well.

## Suggested Solutions

Question 8b.i,

$$p(z) = z^4 - 10z^3 + cz^2 + dz + 169$$

coefficients of  $p(z)$  are real

$\therefore$  complex roots occur in conjugate pairs

$\therefore$  other two roots are  $\bar{\alpha} = a - ib$  and  $\bar{\beta} = b - ia$

8b.ii,

$$\text{Sum of zeros} = -\left(\frac{-10}{1}\right) = 10$$

$$a + ib + a - ib + b + ia + b - ia = 10$$

$$2a + 2b = 10$$

$$a + b = 5$$

$$\text{Product of zeros} = \frac{169}{1} = 169$$

$$(\alpha \bar{\alpha})(\beta \bar{\beta}) = 169$$

$$|\alpha|^2 |\beta|^2 = 169$$

$$|\alpha| |\beta| = 13$$

$$\sqrt{a^2 + b^2} \cdot \sqrt{a^2 + b^2} = 13$$

## Marker's Feedback

8b.i, 8c.ii, Mostly done well. However some students lost marks because of wrong factorisation.

## Suggested Solutions

Question 8b(i), continued

$$a^2 + b^2 = 13$$

$$b = 5 - a \Rightarrow a^2 + (5 - a)^2 = 13$$

$$\therefore 2a^2 - 10a + 12 = 0$$

$$a^2 - 5a + 6 = 0$$

$$(a - 2)(a - 3) = 0$$

$$a = 2 \text{ or } 3; \quad b = 3 \text{ or } 2$$

(8c)

$$p(z) = z^3 + 6z - 4\sqrt{2}i$$

$$p'(z) = 3z^2 + 6$$

$$0 = 3z^2 + 6$$

$$z = \pm \sqrt{2}i$$

$$\begin{aligned} p(\sqrt{2}i) &= (\sqrt{2}i)^3 + 6(\sqrt{2}i) - 4\sqrt{2}i \\ &= -2\sqrt{2}i + 6\sqrt{2}i - 4\sqrt{2}i \\ &= 0 \end{aligned}$$

## Marker's Feedback

8c. Poorly done! Students to practise similar style questions to improve understanding and application.

## Suggested Solutions

Question 8C Continued

$$\begin{aligned}p(-\sqrt{2}i) &= (-\sqrt{2}i)^3 + 6(-\sqrt{2}i) - 4\sqrt{2}i \\&= 2\sqrt{2}i - 6\sqrt{2}i - 4\sqrt{2}i \\&\neq 0\end{aligned}$$

$\therefore \sqrt{2}i$  is a root.

$$\therefore \alpha = \sqrt{2}i$$

$$\begin{aligned}(z - \alpha)^2 &= z^2 - 2(\sqrt{2}i)z + (\sqrt{2}i)^2 \\&= z^2 - 2\sqrt{2}zi - 2\end{aligned}$$

$$z^3 + 6z - 4\sqrt{2}i = (z^2 - 2\sqrt{2}zi - 2)(z + 2\sqrt{2}i)$$

$$\therefore \beta = -2\sqrt{2}i$$

Marker's Feedback

## Suggested Solutions

Question 9a

$$\begin{aligned}\vec{BA} &= z_1 - z_2 \\ &= 1+i - a-bi \\ &= 1-a+i(1-b)\end{aligned}$$

$$\begin{aligned}\vec{BC} &= z_3 - z_2 \\ &= 7+3i - a-bi \\ &= 7-a+i(3-b)\end{aligned}$$

$$i \times \vec{BC} = \vec{BA}$$

$$i \times (7-a+i(3-b)) = 1-a+i(1-b)$$

$$i(7-a) - (3-b) = 1-a+i(1-b)$$

By equating real and imaginary parts

$$-(3-b) = 1-a$$

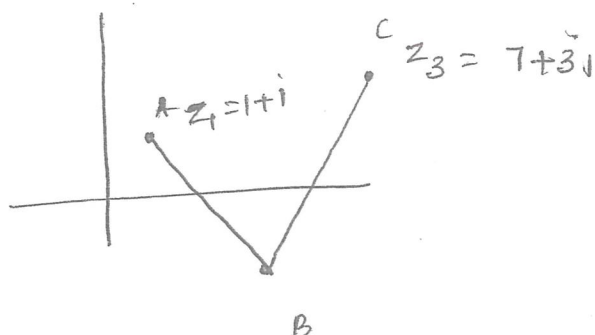
$$b = 4-a \quad \text{and}$$

$$(7-a) = (1-b)$$

$$7-a = 1-(4-a)$$

$$7-a = -3+a$$

$$a = 5$$



$$b = 4-a$$

$$b = 4-5$$

$$b = -1$$

$$z_2 = 5-i$$

Marker's Feedback

Mostly done well.

## Suggested Solutions

Question 9b(i),

$$\therefore |z| = |w|$$

$$\left| \frac{z}{w} \right| = 1$$

$$\begin{aligned} \arg\left(\frac{z}{w}\right) &= \arg(z) - \arg(w) \\ &= \frac{\pi}{3} \end{aligned}$$

9b(ii),

$$z = w \operatorname{cis} \frac{\pi}{3}$$

$$z^3 = w^3 \operatorname{cis} \pi$$

$$z^3 = -w^3$$

$$\therefore z^3 + w^3 = 0$$

done well.

## Marker's Feedback

## Suggested Solutions

Question

9C(i),

$$|z| = \left| \frac{1+i}{1-i} \right| = \frac{|1+i|}{|1-i|} = \frac{\sqrt{2}}{\sqrt{2}} = 1$$

$$|w| = \left| \frac{\sqrt{2}}{1-i} \right| = \frac{|\sqrt{2}|}{|1-i|} = \frac{\sqrt{2}}{\sqrt{2}} = 1$$

9C(ii),

$$\begin{aligned} \arg(z) &= \arg\left(\frac{1+i}{1-i}\right) = \arg(1+i) - \arg(1-i) \\ &= \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2} \end{aligned}$$

$$\begin{aligned} \arg(w) &= \arg\left(\frac{\sqrt{2}}{1-i}\right) = \arg(\sqrt{2}) - \arg(1-i) \\ &= 0 - \left(-\frac{\pi}{4}\right) \\ &= \frac{\pi}{4} \end{aligned}$$

done well

Marker's Feedback

Suggested Solutions

Question 9ciii,

The Quadrilateral OABC is a parallelogram with equal sides, hence it is a rhombus

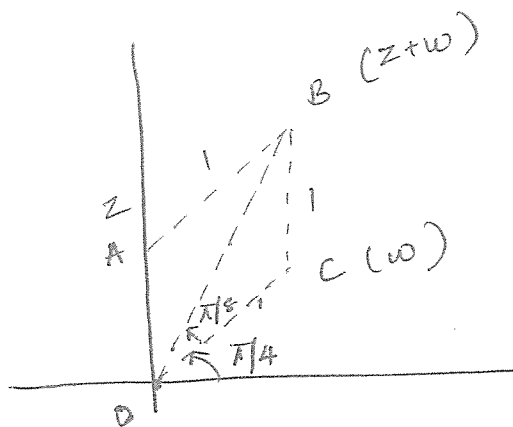
OB is the  $\angle$  bisector of  $\angle AOC$

So,

$$\angle AOC = \frac{\pi}{4}$$

$$\angle AOB = \angle BOC = \frac{\pi}{8}$$

$$\begin{aligned} \text{So the arg}(z+w) &= \frac{\pi}{4} + \frac{\pi}{8} \\ &= \frac{3\pi}{8} \end{aligned}$$



Now,

$$z+w = \frac{1+i}{1-i} + \frac{\sqrt{2}}{1-i}$$

Marker's Feedback

## Suggested Solutions

Question 9C(iii), continued

$$= \frac{(1+\sqrt{2}) + i}{1-i}$$

$$= \frac{(1+\sqrt{2}+i)(1+i)}{(1-i)(1+i)}$$

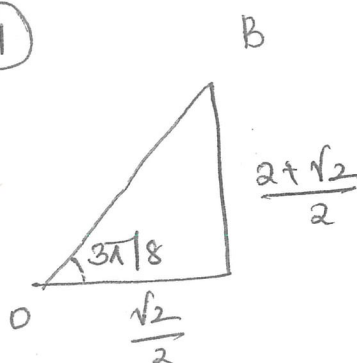
$$= \frac{1+\sqrt{2} + (1+\sqrt{2})i + i - 1}{2}$$

$$= \frac{\sqrt{2} + (2+\sqrt{2})i}{2}$$

$$= \frac{\sqrt{2}}{2} + \frac{2+\sqrt{2}}{2}i \quad (1)$$

$$\tan \frac{3\pi}{8} = \frac{\left(\frac{2+\sqrt{2}}{2}\right)}{\left(\frac{\sqrt{2}}{2}\right)}$$

$$= \frac{2+\sqrt{2}}{\sqrt{2}}$$



$$\tan \frac{3\pi}{8} = \frac{2}{\sqrt{2}} + 1 = 1 + \sqrt{2} \text{ as required.}$$

## Marker's Feedback

Many didn't get the full mark for this question.

## Suggested Solutions

Question

10a i,

$$z = \cos \theta + i \sin \theta$$

$$z^n = \cos n\theta + i \sin n\theta$$

$$\frac{1}{z^n} = z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$\begin{aligned} z^n + \frac{1}{z^n} &= (\cos n\theta + i \sin n\theta) + \cos(-n\theta) + i \sin(-n\theta) \\ &= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ &= 2 \cos n\theta \end{aligned}$$

## Marker's Feedback

Very well done!

## Suggested Solutions

Question 10a, ii,

$$z^5 + \frac{1}{z^5} = 2 \cos 5\theta$$

$$\left(z + \frac{1}{z}\right)^5 = z^5 + 5z^3 + 10z + \frac{10}{z} + \frac{5}{z^3} + \frac{1}{z^5}$$

$$(2 \cos \theta)^5 = \left(z^5 + \frac{1}{z^5}\right) + 5\left(z^3 + \frac{1}{z^3}\right) + 10\left(z + \frac{1}{z}\right)$$

$$32 \cos^5 \theta = 2 \cos 5\theta + 10 \cos 3\theta + 20 \cos \theta$$

$$\cos^5 \theta = \frac{1}{16} (\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta)$$

## Marker's Feedback

Mostly well done but some students forgot coefficients.

## Suggested Solutions

Question 10b(i)

$$(\cos \theta + i \sin \theta)^5 = \cos^5 \theta + 5i \cos^4 \theta \sin - 10 \cos^3 \theta \sin^2 - 10i \cos^2 \theta \sin^3 + 5 \cos \theta \sin^4 + i \sin^5$$

$$(\cos 5\theta + i \sin 5\theta) = (\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta) + i (5 \cos^4 \theta \sin - 10 \cos^2 \theta \sin^3 + \sin^5 \theta)$$

Now equating real and imaginary parts.

$$\cos 5\theta = \cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta$$

and

$$\sin 5\theta = 5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta$$

## Marker's Feedback

Mostly done well again some students forgot coefficients

## Suggested Solutions

Question 10b(ii)

$$\tan 5\theta = \frac{\sin 5\theta}{\cos 5\theta}$$

$$= \frac{5 \cos^4 \theta \sin \theta - 10 \cos^2 \theta \sin^3 \theta + \sin^5 \theta}{\cos^5 \theta - 10 \cos^3 \theta \sin^2 \theta + 5 \cos \theta \sin^4 \theta}$$

$$\frac{\div \cos^5 \theta}{\div \cos^5 \theta}$$

and let  $\tan \theta = t$ 

$$\therefore \tan 5\theta = \frac{5 \tan \theta - 10 \tan^3 \theta + \tan^5 \theta}{1 - 10 \tan^2 \theta + 5 \tan^4 \theta}$$

$$= \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4}$$

as required.

## Marker's Feedback

Very well done, but some students forgot to put it in 't' form but no marks were deducted.

## Suggested Solutions

Question

10b iii,

$$\text{let } \theta = \frac{\pi}{5} \quad \text{so } t = \tan \frac{\pi}{5} \quad \text{and } \tan 5\theta = \tan \pi = 0$$

$$\text{so } \frac{5t - 10t^3 + t^5}{1 - 10t^2 + 5t^4} = 0$$

$$t(t^4 - 10t^2 + 5) = 0$$

$$\text{But } t = \tan \frac{\pi}{5} \neq 0$$

$$(t^2)^2 - 10t^2 + 5 = 0$$

$$\therefore t^2 = \frac{10 \pm \sqrt{100 - 20}}{2}$$

$$t^2 = 5 \pm 2\sqrt{5}$$

but  $\tan x$  is always increasing,  $\tan \frac{\pi}{5} < \tan \frac{\pi}{4}$   
 $\tan \frac{\pi}{5} < 1$  so  $t^2 < 1$

$$t = \pm \sqrt{5 - 2\sqrt{5}}$$

$$\text{but } t = \tan \frac{\pi}{5} > 0$$

$$\therefore \tan \frac{\pi}{5} = \sqrt{5 - 2\sqrt{5}}$$

## Marker's Feedback

Some students let  $\theta = \frac{\pi}{25}$  because of which they could succeed.

Few students got the wrong roots due to incorrect use of Quadratic formula

Students are advised to reason why  $\tan \frac{\pi}{5} = +\sqrt{5 - 2\sqrt{5}}$  otherwise well done!

## Suggested Solutions

Question

10C1,

$$(1 + i \tan \theta)^n + (1 - i \tan \theta)^n$$

$$\left( \frac{\cos \theta}{\cos \theta} + \frac{i \sin \theta}{\cos \theta} \right)^n + \left( \frac{\cos \theta}{\cos \theta} - \frac{i \sin \theta}{\cos \theta} \right)^n$$

$$= \left( \frac{\cos \theta + i \sin \theta}{\cos \theta} \right)^n + \left( \frac{\cos \theta - i \sin \theta}{\cos \theta} \right)^n$$

$$= \frac{\text{cis } n\theta + \text{cis } (-n\theta)}{\cos^n \theta}$$

$$= \frac{2 \cos n\theta}{\cos^n \theta}$$

Marker's Feedback

Mostly well done.

## Suggested Solutions

Question 10c(ii),

$$\text{let } z = i \tan \theta$$

$$(1 + i \tan \theta)^2 + (1 - i \tan \theta)^2 = \frac{2 \cos 2\theta}{\cos^2 \theta}$$

$$\therefore 2 \cos 2\theta = 0$$

$$\therefore 2\theta = \frac{\pi}{2} \text{ or } \frac{3\pi}{2}$$

$$\therefore \theta = \frac{\pi}{4} \text{ or } \frac{3\pi}{4}$$

$$\therefore z_1 = i \tan \frac{\pi}{4} \quad ; \quad z_2 = i \tan \frac{3\pi}{4} \\ = i \tan \left( -\frac{\pi}{4} \right)$$

$$\therefore z = \pm i \tan \frac{\pi}{4}$$

## Marker's Feedback

Poorly done. Most students did not know how to use 10c(i), to solve 10c(ii),

## Suggested Solutions

## Question 11 a.i,

Roots of  $z^5 - 1 = 0$  are

$$\cos \frac{2\pi}{5}; \cos \frac{4\pi}{5}; \cos \frac{6\pi}{5}; \cos \frac{8\pi}{5}$$

$$\omega = \cos \frac{2\pi}{5}$$

$$\omega^2 = \cos \frac{4\pi}{5} \rightarrow \text{which is a root of } z^5 - 1 = 0$$

$$\omega^3 = \cos \frac{6\pi}{5}$$

$$\omega^4 = \cos \frac{8\pi}{5}$$

11 a.ii,

$$\text{Sum of roots: } \alpha + \beta + \gamma = -\frac{b}{a}$$

Roots of  $z^5 - 1 = 0$  add upto '0'

$$\therefore 1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$$

$$\omega + \omega^2 + \omega^3 + \omega^4 = -1$$

Alternatively as  $\omega$  is a root of  $z^5 - 1 = 0$ ,

$$(\omega - 1)(\omega^4 + \omega^3 + \omega^2 + \omega + 1) = 0$$

$$\text{As } \omega \neq 1, \omega^4 + \omega^3 + \omega^2 + \omega + 1 = 0 \Rightarrow \omega^4 + \omega^3 + \omega^2 + \omega = -1$$

## Marker's Feedback

Generally OK. This was leniently marked.

$(\omega^2)^5 = (\omega^5)^2 = 1$  was also allowed.

In Part (ii) when students used  $(\omega - 1)(\omega^4 + \omega^3 + \omega^2 + \omega + 1) = 0$  they often did NOT mention  $\omega \neq 1$ .

## Suggested Solutions

Question 11 b, i,

$$z^5 - 1 = (z - 1) \left( z - \operatorname{cis} \frac{2\pi}{5} \right) \left( z - \operatorname{cis} \frac{8\pi}{5} \right) \left( z - \operatorname{cis} \frac{4\pi}{5} \right) \left( z - \operatorname{cis} \frac{6\pi}{5} \right)$$

Now  $\left( z - \operatorname{cis} \frac{2\pi}{5} \right) \left( z - \operatorname{cis} \frac{8\pi}{5} \right)$

$$= z^2 - z \left( \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{8\pi}{5} + i \sin \frac{8\pi}{5} \right) + 1$$

$$\left( \because \cos \frac{8\pi}{5} = \cos \frac{2\pi}{5} \right)$$

$$= z^2 - z \left( \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{2\pi}{5} - i \sin \frac{2\pi}{5} \right) + 1$$

$$\left( \because \sin \frac{8\pi}{5} = -\sin \frac{2\pi}{5} \right)$$

$$= z^2 - 2z \cos \frac{2\pi}{5} + 1$$

Hence

$$\therefore z^5 - 1 = (z - 1) \left( z^2 - 2z \cos \frac{2\pi}{5} + 1 \right) \left( z^2 - 2z \cos \frac{4\pi}{5} + 1 \right)$$

## Marker's Feedback

Generally well done. A small minority of students forgot to write the first line  $(z - \operatorname{cis} \frac{2\pi}{5})(z - \operatorname{cis} \frac{4\pi}{5}) \dots \text{etc.} = z^5 - 1$ .

## Suggested Solutions

Question 11b(i),

Hence

$$\text{as } z^5 - 1 = (z-1)(z^4 + z^3 + z^2 + z + 1)$$

$$(z \neq 1) (z^2 - 2z \cos \frac{2\pi}{5} + 1) (z^2 - 2z \cos \frac{4\pi}{5} + 1) = (z \neq 1) (z^4 + z^3 + z^2 + z + 1)$$

$$\therefore (z^2 - 2z \cos \frac{2\pi}{5} + 1) (z^2 - 2z \cos \frac{4\pi}{5} + 1) = z^4 + z^3 + z^2 + z + 1$$

11b(ii),

$$z^4 + z^3 + z^2 + z + 1 = (z^2 - 2z \cos \frac{2\pi}{5} + 1) (z^2 - 2z \cos \frac{4\pi}{5} + 1)$$

$$= (z^2 - 2z \cos \frac{2\pi}{5} + 1) (z^2 + 2z \cos \frac{\pi}{5} + 1)$$

$$(\text{as } \cos \frac{4\pi}{5} = -\cos \frac{\pi}{5})$$

By dividing both sides by  $z^2$  and rearranging

$$\left(z^2 + \frac{1}{z^2}\right) + \left(z + \frac{1}{z}\right) + 1 = \left(z + \frac{1}{z} - 2 \cos \frac{2\pi}{5}\right) \left(z + \frac{1}{z} + 2 \cos \frac{\pi}{5}\right)$$

$$\text{Sub } z = e^{i\theta}; \quad z^2 + \frac{1}{z^2} = 2 \cos 2\theta \quad \text{and} \quad z + \frac{1}{z} = 2 \cos \theta$$

$$2 \cos 2\theta + 2 \cos \theta + 1 = (2 \cos \theta - 2 \cos \frac{2\pi}{5}) (2 \cos \theta + 2 \cos \frac{\pi}{5})$$

## Marker's Feedback

(b)(i) Well done.

(b)(ii) Students either realised to  $\div z^2$  and solved the entire question or could not complete the question if they didn't.Many students didn't explain that  $\cos \frac{4\pi}{5} = -\cos \frac{\pi}{5}$ .

## Suggested Solutions

Question 11iv,

$$\left(2\cos\theta - 2\cos\frac{2\pi}{5}\right)\left(2\cos\theta + 2\cos\frac{\pi}{5}\right) = 2\cos 2\theta + 2\cos\theta + 1$$

Substituting  $\theta = \frac{\pi}{3}$

$$\left(2\cos\frac{\pi}{3} - 2\cos\frac{2\pi}{5}\right)\left(2\cos\frac{\pi}{3} + 2\cos\frac{\pi}{5}\right) = 2\cos\frac{2\pi}{3} + 2\cos\frac{\pi}{3} + 1$$

$$\left(1 - 2\cos\frac{2\pi}{5}\right)\left(1 + 2\cos\frac{\pi}{5}\right) = -2 + 2 + 1$$

$$\left(1 - 2\cos\frac{2\pi}{5}\right)\left(1 + 2\cos\frac{\pi}{5}\right) = 1$$

## Marker's Feedback

(iv) Most students realised to substitute  $\theta = \frac{\pi}{3}$  & completed the question well.

A small minority tried the wrong angle & could not complete the question.